General Care And Maintenance

1. Plug/Unplug cables(excluding USB cables), connectors or any connected hardware(RPI Camera inclusive) only when power supply is OFF
   - Failure to do so may result in hardware failure or short circuit

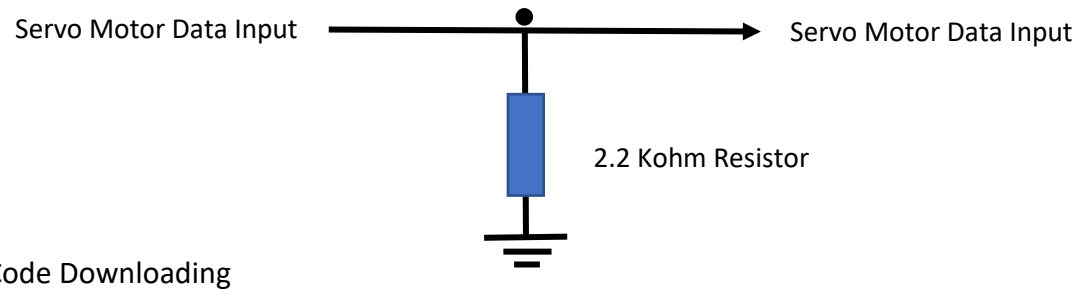
2. Do not leave PCB or Bare wires dangling or uninsulated
   - Bare wires from connections made from such as jumper cables, resistors, sensor cables, must be insulated with electrical tape. **Insulation Tape** can be obtained from respective Labs. **DO NOT USE BLUTACK as they can short components too**
   - The underside of the PCB(Raspberry/STM32/RPI Camera etc) contains hundreds of bare components which are electrical conducting. They could potentially short together when bare wires or touched on metal surfaces.

3. Secure all PCBs onto the Nanocar using the metal studs provided.

4. Hold PCB by the edges. Do not touch any ICs on the PCB
   - Static electricity could accumulate on your hand, resulting electrostatic discharge (ESD). This discharge will kill a chip.
   - The most proper way is to have an ESD Wrist strap on your wrist and grounded properly when handling PCBs

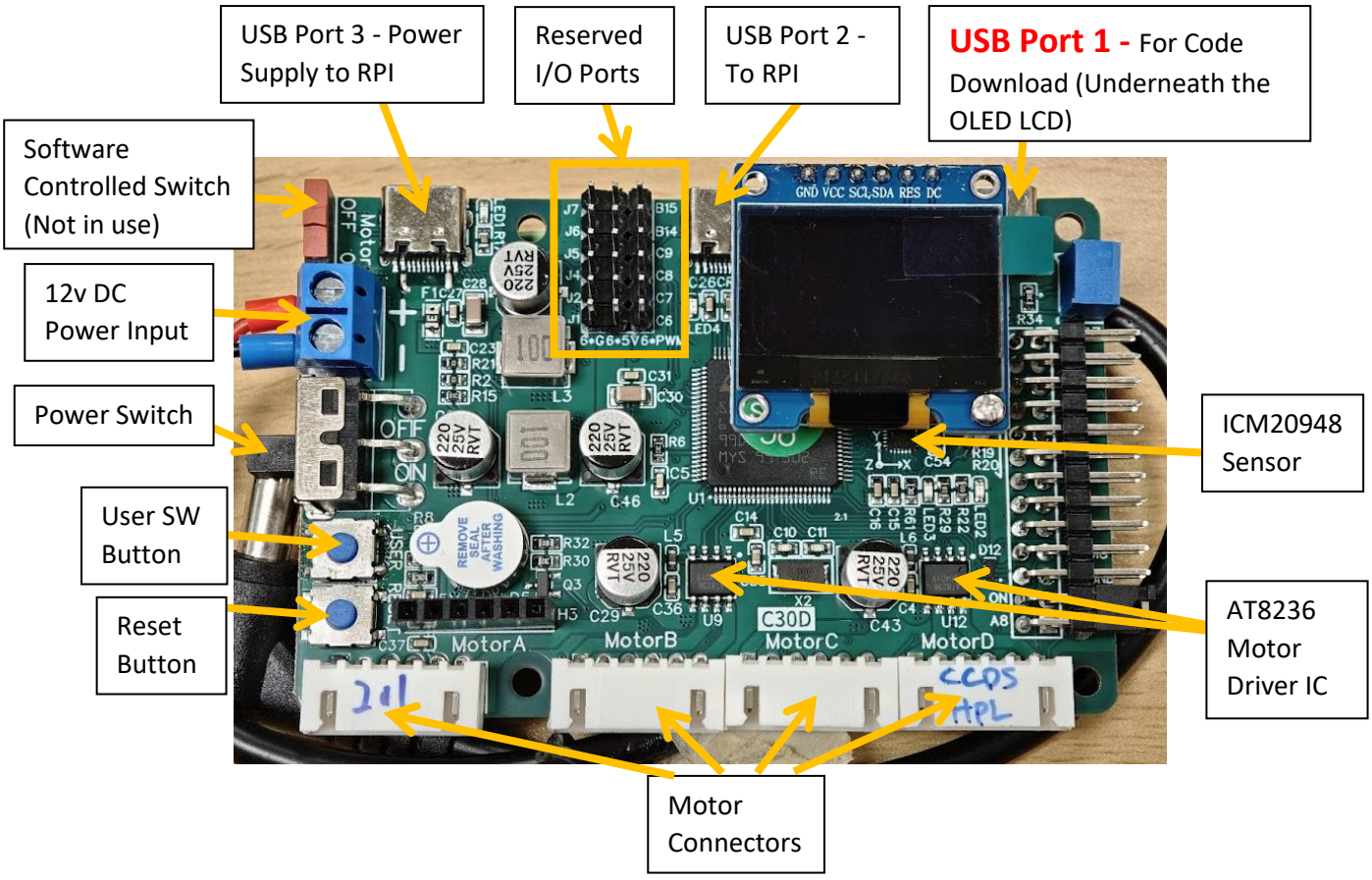
STM32F103VET6 C30D Ver2.1 Wheeltec Board

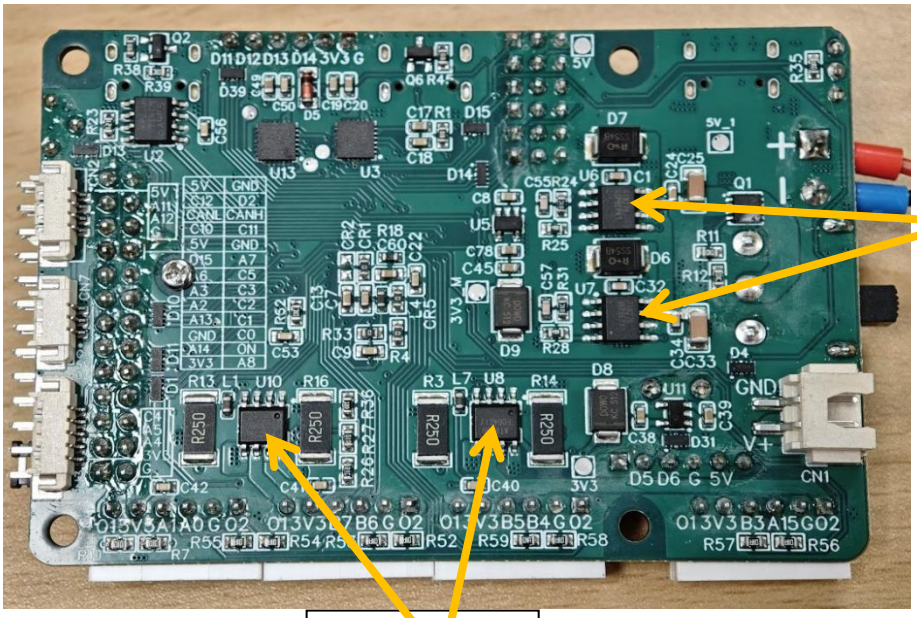
1. Hardware Modification (Optional but for those confident with their soldering skill, try this as it should help with the random twitching and huge current drawn during start up transient state)
   (To be verified), Apply a pull down resistor to the input of the servo motor. This prevents the servo from moving randomly and damage the board

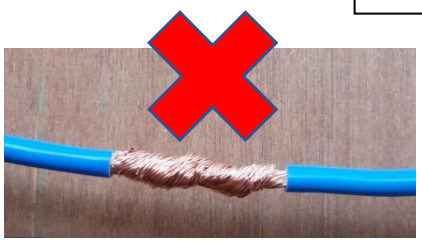


Code Downloading

- Power off the battery supply by switching off the big slider switch
- ***Reminder* Do not turn on the battery power when USB cable is in Port 1**
- Disconnect the servo motor power from the reserved I/O port. The servo motor potentially will draw a very high current during transient states (When USB cable is inserted, damaging the STM32 board in the process
- Connect USB to port 1, USB Board underneath the OLED LCD) of the board,
- Download the code
- After downloading, Unplug the USB cable from port 1


USB Port 3 - Power Supply to RPI
Reserved I/O Ports
USB Port 2 - To RPI
USB Port 1 - For Code Download (Underneath the OLED LCD)
Software Controlled Switch (Not in use)
12v DC Power Input
Power Switch
User SW Button
Reset Button
ICM20948 Sensor
AT8236 Motor Driver IC
Motor Connectors
MotorA
MotorB
MotorC
MotorD


5v Step Down Converter IC RT8279 (for STM32 & RPI)
AT8236 Motor Driver IC




Insulating Tape

